Lecture 6: Law of Large Numbers, Central Limit Theorem, & Correlation

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1. Agenda and Announcements

- Today

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- Hypothesis Testing and Confidence Intervals
- Top Hat Quiz
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- Next class

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    - Lab 3 completion
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1.1

- Announcements

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    - CASA Registration
    - No class next Monday or Tuesday (March 10)
    - I will record a midterm review and post an online pre-midterm quiz
**for points**
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2. Hypothesis testing

2.1 What is a hypothesis

A falsifiable statement about what we believe will happen based on the theory we are trying to test.

2.2 Falsifiability

- We start from the assumption that our theory is wrong
- Our hypothesis is called the

alternative hypothesis or **H1**

- because it is the alternative to our assumption of no relationship

null hypothesis or **H0**

2.3 Testing the null hypothesis

- Statistical tests show the degree of certainty that we can reject the null hypothesis
- They show the probability that the alternative hypothesis is due to random chance
- When the probability, *p*, is below our pre-determined threshold, we reject the null hypothesis
- **If we reject the null hypothesis, the alternative hypothesis is true, right?**

2.4 NO!

- **If we reject the null hypothesis, the alternative hypothesis is true, right?**

NO!!!!

- If we reject the null, we infer that the alternative hypothesis (our hypothesis) is approximately true within the probability we have chosen.
- If we reject the null, we can say there is strong evidence that the alternative hypothesis (our hypothesis) is true
- We often test multiple hypotheses

2.5 How do we get there?

How do we get from a sample with a correlation to talking about testing a hypothesis for a population?

- Tying sample statistics to population parameters (LLN)
- Probability distributions
- Tying our data to the probability distributions (CLT)

2.6 Statistics to parameters

- We rely on the Law of Large Numbers and use sample statistics
- With sufficient sample size and proper math, our sample mean is an unbiased estimator of the population mean
- With CLT, s^2 is an unbiased and consistent estimator of σ^2
- For the normal distribution s^2 is also the Minimum Variance Unbiased Estimator for σ^2

2.7 Probability distributions

The *68-95-99.7 Rule*

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            + Allows us to estimate probability based on distance from the
mean
            + Applies to normal distribution
            + Basis for the actual decision rules
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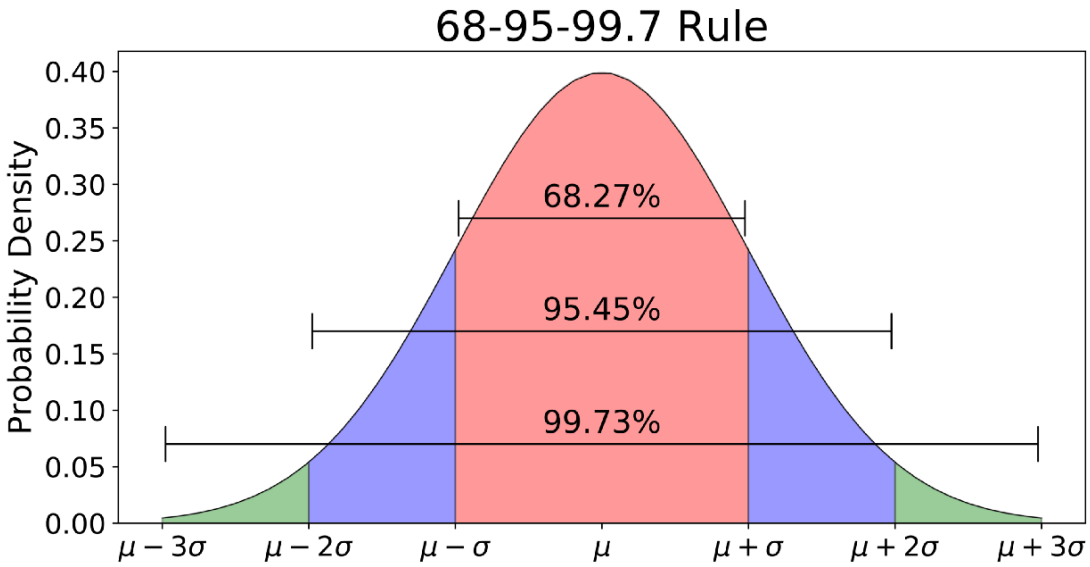
2.8 The 68-95-99.7 Rule



68-95-99.7 rule

2.9 From Distribution to Confidence

- We can now bridge the gap between probability distributions and our sample data.
- CLT ties our sample means to the normal distribution
- Sample stats are unbiased estimators of population parameters
- They are only *point estimates*, so we construct a *confidence interval*
- *confidence interval* a range of values around the *point estimate* that we are X% confident contains the true population parameter

2.10 Math and philosophy of the confidence interval

- Math and philosophy detail: The true population parameter is a fixed but unknown number, while our calculated interval is random but known with certainty. Our X percent confidence reflects the long-run reliability of our method. If we repeated this exact sampling process 100 times, X of the resulting intervals would successfully capture the true population parameter.

2.11 Calculating the Interval

- we need three pieces of information:

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  - point estimate = sample statistics
  - standard error = the standard deviation of the sampling
distribution
  - critical value derived from our desired confidence level - for
normal distribution this is a z-score
  - For 95% confidence, the critical z is 1.96
```

- The general formula for a confidence interval is:

$$CI = \bar{x} \pm z\left(\frac{s}{\sqrt{n}}\right)$$

CI of the variance

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$$

- This use the Chi Square distribution which we will discuss after midterm

2.12 CI of the standard deviation

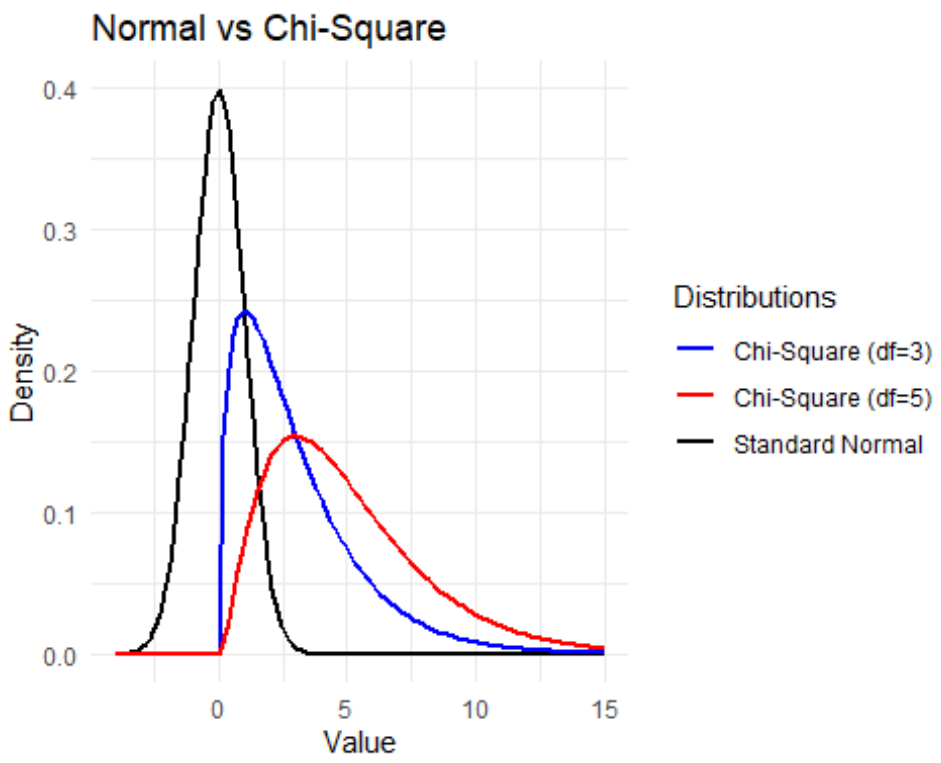
$$\sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha/2}}} \leq \sigma \leq \sqrt{\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}}$$

- Also uses the Chi-square distribution

2.13 Why the Chi-Square not the normal?

- When we square distances from the mean to get the variance, the distribution changes
- By eliminating all negative numbers, the distribution becomes skewed and looks like this:

2.14 Distribution Comparison



2.15 Authorship, License, Credits

- Stephen Moore code used to simulate the Law of Large Numbers
- 68-95-99 rule graphic Source:https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2
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- Website: tomhanna.me
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